

Zero injection solutions for on-grid inverters

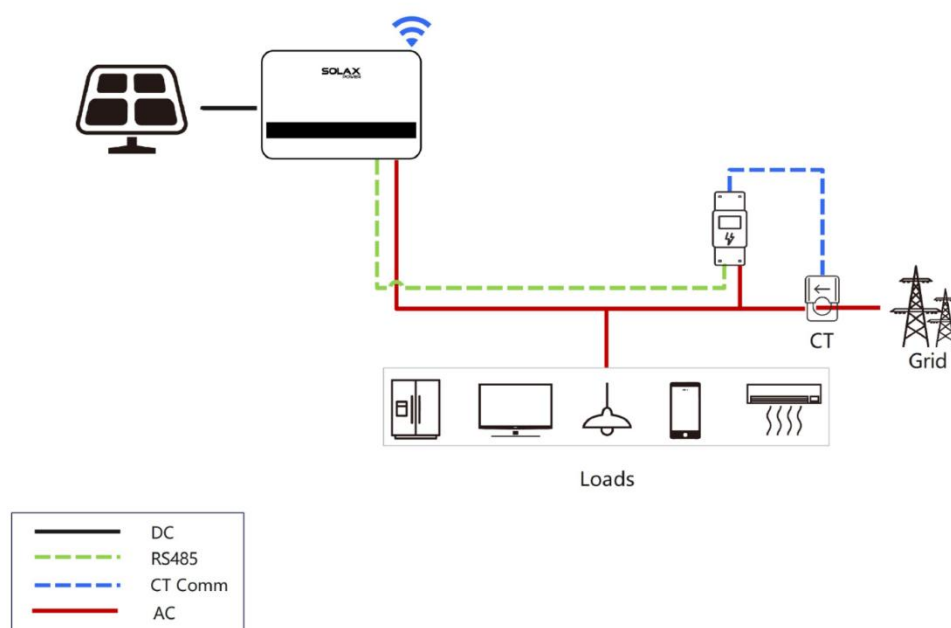


Nowadays, photovoltaic systems have been widely used all around the world. There are plenty strengths of using solar power to supply loads, reducing carbon emissions, lowering electricity costs, and even selling electricity to the grid for profit. In a solar system, the inverter can convert the DC power to AC power, which supplies the load first and the excess will be injected into the grid. However, due to the need to maintain grid quality, not all countries or electricity companies can allow generators to export to the grid, so the need to control zero output has arisen.

For now, Solax has released three solutions to control zero injection, which are built-in export control function, power bias function, and per phase control function.

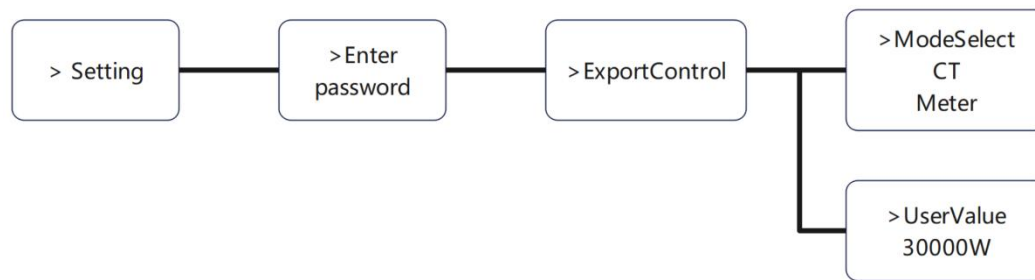
1. Built-in export control

This is a basic export control function integrated in all Solax inverters, which need to be used with a measuring device: Meter, Meter with CT or standalone CT. The following diagram shows the wiring of a meter with CT, both the AC and communication lines need to be connected to ensure accurate export control, for the specific connection steps please refer to the article: <https://kb.solaxpower.com/solution/detail/ff8080818407e2a701840a15af480028>



The meter detects the consumption power and reports it to the inverter, then based on the integrated export control algorithm, the inverter can limit the output to only be sufficient to the load, so as to achieve zero injection to the grid. After the system wiring mentioned in the above articles, system configuration is also required. Users can set up the export control function locally via LCD or remotely via APP and

monitoring platform, taking the local setting as an example, the steps are as follows:



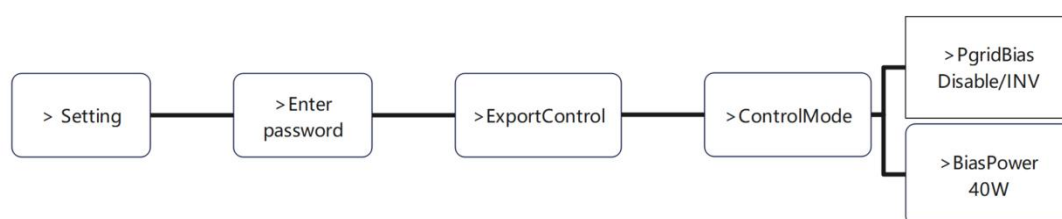
Note: The 'CT' option is not available for three-phase on-grid inverters, which can only work with meter or meter with CT at the moment.

Users can set the power injected to the grid according to how much power is required as 'User Value' option and set it to zero if zero export is needed.

2. Power bias

Due to light power fluctuations caused by load changes during export control, we have developed a new function Power Bias on the basis of the built-in export control function, thus maximizing the absence of any power output.

Power bias is to prefer to buy electricity from the grid rather than an injection, which is mainly suitable for sites with strict restrictions on zero export. Users can enable this function as needed as follows:



Users need to enable the power bias function by choosing the 'INV' item, and if needed, the bias power can be set. This function now has released on the following models, and the default value and its defining range can be found in the table below:

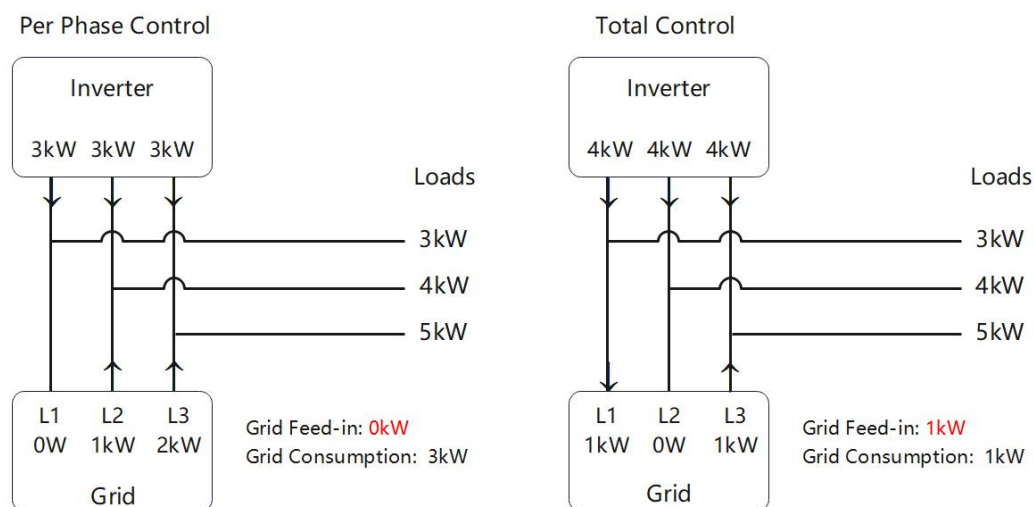
Model	Default Value	Range
X1-MINI G4	40W	1~100W
X1-BOOST G4	40W	1~100W
X3-MIC G2	40W	1~300W

X3-PRO G2	40W	1~300W
X3-MEGA G2	10%Pn	0~10%Pn
X3-FORTH	10%Pn	0~10%Pn

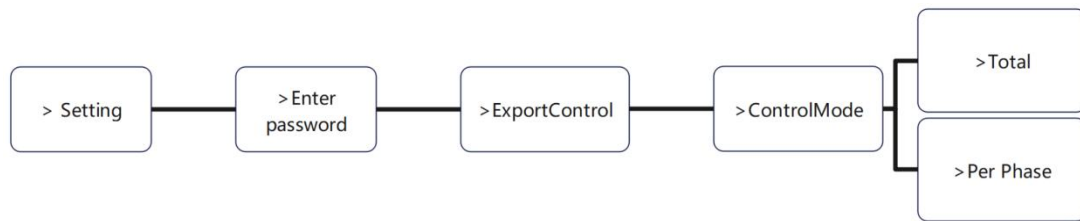
3. Per phase control

For single-phase inverters, the above two solutions are sufficient for accurate control of the zero output. However, for three-phase inverters, since nonsupport three-phase unbalanced output, according to the existing logic, the inverter will control the zero output by the way of calculating the average of the power of three phases' loads, which is called 'Total control'. But in this case, it may happen that one phase has power output when the loads on the three phases are different. To prevent this situation, we have developed a function to control the output for each phase, and we call it 'Per phase control'.

Per phase control means that the phase with the smallest load is used as the reference, and each phase of the inverter outputs at the power of the reference, resulting in an overall zero grid injection. The following diagram shows the difference between the total control and per phase control, which demonstrates how per phase control achieves the complete zero output:



As can be seen from the diagram, with different loads, there is a possibility that one phase is on output when using total control, whereas the per-phase control solution would not. To enable the per phase control, users can configure the inverter as follows:

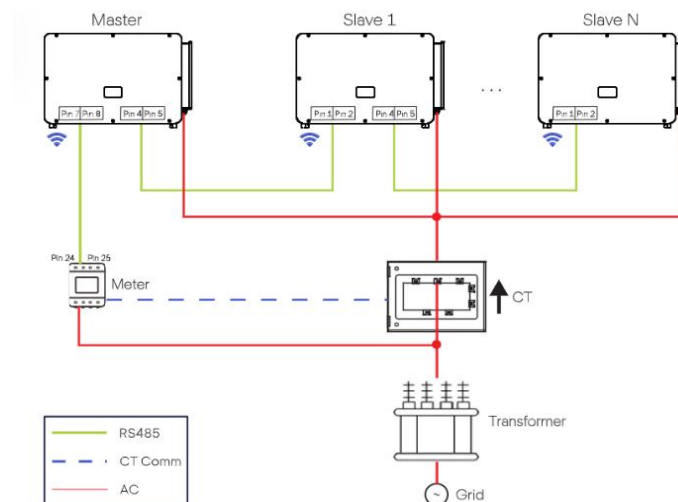


Note: For three phase on-grid inverters, users should choose the export control mode first, and then set up the power bias function.

4. Parallel Connection

For Modbus

Take the Forth machine as an example, a 485 communication wire must be attached directly to the inverter. The devices are connected in a daisy chain type connection mode. The Master's RS485-2 is connected to the electricity meter, and the Master and slave are connected to the RS485 1 port. The bus topology is shown as follows:



Set parallel system with Modbus function The devices are connected in a bus type connection mode. The Master's RS485-2 is connected to the electricity meter, and the Master and slave are connected to the RS485 1 port. The slave device needs to set its Modbus address and baud rate. Set the Modbus address of the slave device on the power station to 2-11 (up to 10 machines are supported at present) and the baud rate to 9600.

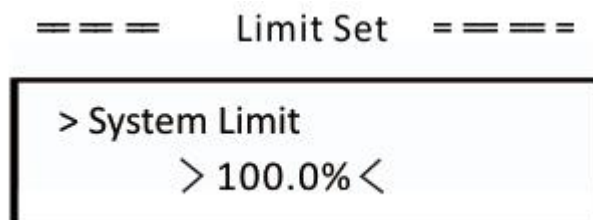
= Communication Parameter =

> Modbus Addr
2

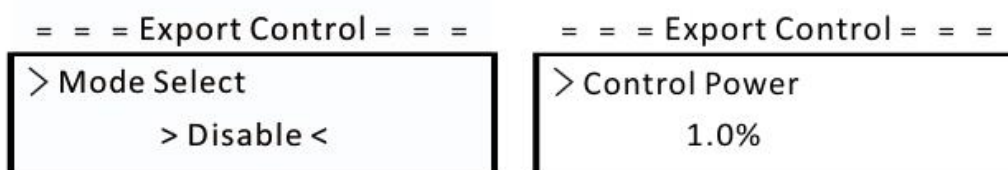
= Communication Parameter =

> Modbus Baud
9600

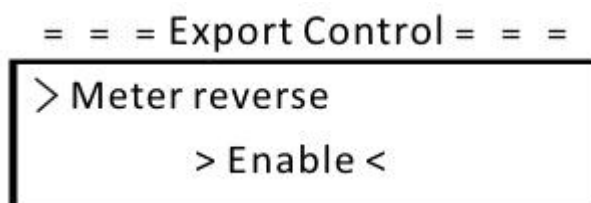
The equipment connected to the meter is selected as the Master, and the Master mode and anti-reflux function of the Master equipment on the power station are enabled by APP/ web page/screen, among which System Limit are set to 100%



With power control function, the inverter can control the energy exported to the grid. The "Control Power" can be set by the installer. When you set 100% for control power, it means the energy can be exported to grid with full power. When you set 0%, Exporting to grid is limited. Please set the percentage according to the actual need. Choose "Disable" means the function will not be activated. Press "Up" and "Down" button to select and press "Enter" to confirm it.

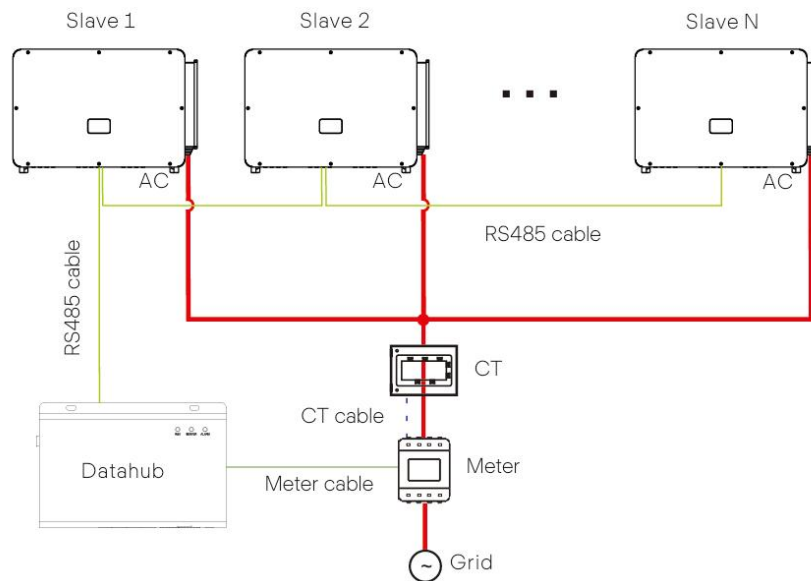


Installer can set "Soft Limit" and "Hard Limit" for export control.

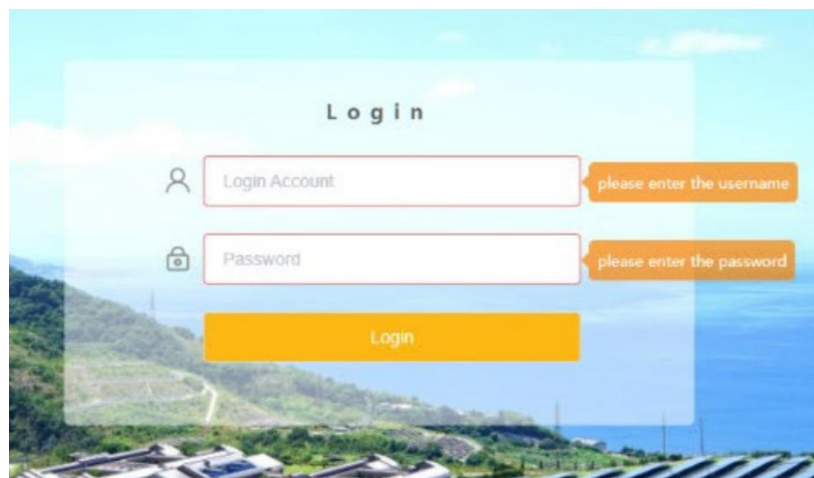


For Datahub

The series inverter provides the parallel connection function when connected with Datahub, which could support at most 60 inverters to parallel in one system and can control zero injection to the grid with a meter installed in the main circuit. In this parallel system, the Datahub will be the master of the system, and all the inverters are the slaves. The Datahub can communicate with all the slave inverters.



Connect your computer to the DataHub hotspot WiFi with the WiFi _xxxxxx (DataHub's Serial Number), and then visit 192.168.10.10 to access the login page. Administrator account: admin, initial password: DataHub's Serial Number. User account: user, initial password: 123456. Guest account: visitor, initial password: 123456.



On the homepage, select DataHub Setting > Serial Port Setting to check the baud rate. Make sure the baud rate is consistent with inverters.

SOLAX POWER admin Select Language Log Out

Smart Scene

Device Upgrade

DataHub Setting

Internet Setting

Time Setting

Serial Port Setting

Other Setting

DataHub Info

Serial Port Setting

RS485 Channel	Agreement Type	Baud Rate	Verification Method	Stop Bit
1	modbus	9600	No Verification	1
2	modbus	9600	No Verification	1
3	modbus	9600	No Verification	1
4	modbus	9600	No Verification	1

Save

After accessing the Site Settings, click on the "Export Limit Control" feature.

SOLAX POWER admin Select Language

Overview

Site Management

Site Setting

Export limit Control

Power Control

reactive power compensati...

EV Charger Settings

Electricity Price Setting

Meter Setting

Export Limit Control

Enable Disable ☒ Enable

Control Mode Total

Feed-in Buffer Please Select

Export Limit Control Rapidly Disable ☒ Enable

* Export Power(%) 0.0

Feedin [+] , Consumed [-]

Reset Submit

5. Capacity

In summary, all three of the above options provide good control of the grid connection and the customer can set up the inverter according to the specific conditions of their plant. The correspondence between each grid-connected inverter model and their respective supported functions is as follows:

Model	Built-in Export Control	Power Bias	Total/Per Phase Control
X1-MINI G3	Yes	No	/
X1-MINI G4	Yes	Yes	/
X1-BOOST G3	Yes	No	/
X1-BOOST G4	Yes	Yes	/
X1-SMART G1	Yes	No	/
X3-MIC G1	Yes	Yes	Total / Per Phase

X3-MIC G2	Yes	Yes	Total /Per Phase
X3-PRO G1	Yes	Yes	Total /Per Phase
X3-MAX	Yes	No	Total /Per Phase
X3-PRO G2	Yes	Yes	Total /Per Phase
X3-MEGA G1	Yes	No	Total /Per Phase
X3-MEGA G2	Yes	Yes	Total /Per Phase
X3-FORTH	Yes	Yes	Total /Per Phase

Note: Please update the firmware to the latest version before activating the Power Bias function and Per Phase function.